

Algorithmic Pseudocode Sample 53

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1: possible pair list  $\leftarrow$  conventional neighbour list
2: new pair list  $\leftarrow \emptyset$ 
3: for all stickers  $i$  do
4:    $N_{bonds}[i] \leftarrow 0$ 
5: end for
6: shuffle possible pair list
7: for all pairs  $(i, j) \in$  possible pair list do
8:    $\Delta E_{ij} \leftarrow U_{bound}(r_{ij}) - U_{unbound}(r_{ij})$ 
9:   if  $(i, j) \in$  previous pair list then
10:    if  $X \sim U(0, 1) > \exp(-\Delta E_{ij})$  then
11:      append  $(i, j)$  to new pair list
12:       $N_{bonds}[i] \leftarrow N_{bonds}[i] + 1$ 
13:       $N_{bonds}[j] \leftarrow N_{bonds}[j] + 1$ 
14:    end if
15:  else if  $N_{bonds}[i] < functionality$  and  $N_{bonds}[j] < functionality$  then
16:    if  $X \sim U(0, 1) < \exp(-\Delta E_{ij})$  then
17:      append  $(i, j)$  to new pair list
18:       $N_{bonds}[i] \leftarrow N_{bonds}[i] + 1$ 
19:       $N_{bonds}[j] \leftarrow N_{bonds}[j] + 1$ 
20:    end if
21:  end if
22: end for
23: previous pair list  $\leftarrow$  new pair list
```
